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PHOTONIC DEVICES HAVING A FIELD-EFFECT TRANSISTOR (FET)

Abstract

Systems and methods are provided for optical devices having a field-effect transistor (FET) for optical tuning. Examples include a device layer formed on a substrate and comprising a first material, and a FET formed on the substrate. The MOSFET comprises a drain formed in the device layer comprising a first doped region, a source formed in the device layer comprising a second doped region, a gate comprising a second material formed on the device layer, and a conductive channel formed in the device layer based on a voltage bias applied to the gate. An optical waveguide is formed between the gate and the conductive channel.

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Background/Summary

BACKGROUND

[0001] Silicon photonic devices can manipulate light for information processing by altering a refractive index of a silicon waveguide within a photonic integrated circuit (PIC). One example method for altering the refractive index is through electro-optic effects, such as the plasma dispersion effect, which varies free-carrier concentration within the silicon waveguide. Varying of the free-carrier concentration can effectively change the index of refraction of the waveguide. Another example approach is metal-oxide-semiconductor (MOS) capacitor-based modulators that can produce carrier accumulation within a waveguide.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] The present disclosure, in accordance with one or more various embodiments, is described in detail with reference to the following figures. The figures are provided for purposes of illustration only and merely depict typical or example embodiments.

[0003] FIGS. **1**A and **1**B illustrate an example optical device in accordance with implementations disclosed herein.

[0004] FIGS. **2**A and **2**B illustrate another example optical device in accordance with implementations disclosed herein.

[0005] FIG. **3** depicts a schematic diagram of an optical device in implemented as a Mach-Zehnder Interferometer in accordance with an example disclosed herein.

[0006] FIG. **4** illustrates an example optical device implemented as a ring resonator in accordance with an example disclosed herein.

[0007] FIG. **5** is an example computing component that may be used to implement various features of optical devices in accordance with the implementations disclosed herein.

[0008] FIG. **6** is an example computer system that may be used to implement various features of optical devices of the present disclosure.

[0009] The figures are not exhaustive and do not limit the present disclosure to the precise form disclosed.

DETAILED DESCRIPTION

[0010] Silicon photonics is an evolving field that finds use in various technological fields, such as, but not limited to, communication applications, computing applications, and sensing applications, as well as many others. As outlined above, silicon photonics have been implemented to manipulate optical signals (e.g., light) for encoding and/or decoding of information for processing, for example, by altering refractive indices of waveguides within a PIC. Generally, the refractive index of a waveguide can be altered by tuning a voltage bias across a silicon photonic device, which changes carrier concentration within the waveguide. Changes in carrier concentration induces a change in refractive index that alters a phase of the waveguide. The amount of change induced in a waveguide may be referred to as “modulation depth”. Smaller modulation depths (e.g., smaller changes in phase) may require increased precision in the tuning and detection in order to accurately encode/decode information on to an optical signal. However, in conventional silicon photonic devices, larger modulation depths generally require larger regions within the silicon photonic device for altering carrier concentration. Increasing this region can reduce the speed of the device due to signals having to traverse a larger region to induce the desired change, and may require an increased footprint within the PIC itself.

[0011] Furthermore, with the advent of memristor photonics, metal oxide-based memristors can be integrated directly onto photonic devices and circuits to provide for non-volatile memory directly on a chip. However, larger voltage biases applied may induce current flow beyond a current limit (also referred to as a current compliance threshold) and exceeding this limit can cause thermal breakdown of the memristor, which can render the device permanently unusable as a memristor. Due to the current limit and a desire to apply as much voltage bias as possible to transition between states of a memristor, it can be difficult to keep the current flowing through such devices below the current limit. Therefore, it may be beneficial to induce current compliance that electrically keeps current in the device below the current limit, which can mitigate thermal break down in order to protect the device.

[0012] The technology of the present disclosure provides optical devices that comprise a MOS field-effect transistor (MOSFET) coupled to a waveguide that can be used for various purposes to overcome the above-described technical shortcomings, as well as provide additional benefits. In some implementations, the optical devices disclosed herein can function as an optical modulator with improved modulation efficiency by providing increased modulation depth, which provides for increased range of modulation. For example, the optical devices disclosed herein can provide multiple structures within the device that can be simultaneously or individually tuned to increase the modulation depth of the devices. In this case, the MOSFET may be one such structure that can be tuned by applying a voltage bias to a gate to enter a saturation region of the MOSFET, which can increase the length of a channel and carrier concentration. A second structure can be tuned to accumulate and/or deplete these carriers to other regions within the device. The increased carrier concentration within the device can translate to increased modulation depth.

[0013] In another implementation, the disclosed optical devices can function as an optical power monitor within an optical waveguide that does not require extracting an optical signal from the waveguide. For example, the optical device disclosed herein can comprise and/or be integrated into a Mach-Zehnder Interferometer (MZI). In this case, the MOSFET can be tuned to monitor optical power of an optical signal in one arm, which can inform the optical power propagating in another arm of the MZI.

[0014] In another example, the disclosed optical devices can also be controlled to limit an amount of current flowing through the device. For example, the optical device may comprise the MOSFET connected in series to another MOS diode (e.g., a memristive structure in various examples). Thus, the MOSFET can be driven by applying a voltage bias at the gate to induce pinch-off of a channel in the MOSFET. The pinch-off can function to limit current flowing through into the memristive structure and, by setting the voltage bias appropriately, can be used to mitigate or avoid thermal breakdown by electrically constraining the current flowing into the memristive device to below a current limit.

[0015] Examples of the technology disclosed herein provide for optical devices, such as silicon photonic devices, which can be integrated into silicon photonic platforms and PICs. Examples disclosed herein include a substrate having a device layer formed thereon. The device layer may comprise a first material, such as but not limited to, silicon or other Group IV material. A MOSFET in accordance with the present disclosure is formed on the substrate. The MOSFET comprises a drain, gate, and source. The drain can be formed of a first doped region in the device layer, the source can be formed of a second doped region in the device layer, and the gate can be formed on the device layer. The gate can comprise a second material, such as but not limited to, a Group III-V material. Examples of Group III-V materials that may be used as the second material include, but are not limited to, Indium Phosphide (InP), Gallium Nitride (GaN), and gallium arsenide (GaAs). In an illustrative example, the second material is GaAs. A conductive channel region can be formed in the device layer (e.g., between the source and drain) based on applying a voltage bias to the gate. An optical waveguide can be formed in the device layer above the conductive channel region, for example, between the conductive channel region and the gate.

[0016] In some examples, the first and second doped regions comprise a first doping polarity. For example, the first material may be doped with a first doping concentration and the first doping polarity and the first and second doped regions may comprise a second doping concentration that is higher than the first doping concentration. The second material may comprise a second doping

polarity that is the opposite of the first doping polarity. For example, the first doping polarity may be p-type doping and the second doping polarity may be n-type doping. In this case, the conductive channel may be a p-type channel. In another example, the doping polarities may be reversed.
[0017] FIGS. **1**A and **1**B illustrate an example optical device **100** in accordance with implementations disclosed herein. FIG. **1**B is a top down view of the optical device **100** and FIG. **1**A is a section view of the optical device **100** taken along a line A-A′ shown in FIG. **1**B. In FIG. **1**B, power sources **136***b* and **136***a* are removed for illustrative purposes only. Optical device **100** may be an example of a PIC that can be provided on a chip. In various examples, optical device **100** may be integrated into a silicon photonics platform.
[0018] Optical device **100** includes a substrate **108** having a device layer **106** formed thereon. The device layer **106** may comprise a first material, such as but not limited to, silicon or other Group IV material. The substrate **108** may be provided as silicon or other Group IV material. Other examples of materials for substrate **108** may include, but are not limited to, Silicon Nitride (Si.sub.3N.sub.4), Aluminum oxide (Al.sub.2O.sub.3), Hafnium Dioxide (HfO.sub.2), diamond, silicon carbide (SiC), or combinations thereof.
[0019] The optical device **100** comprises a MOSFET formed on the substrate **108**. The MOSFET can be formed between a drain **124**, gate **104**, and source **116**. The drain **124** can be formed as a first doped region **123** in the device layer **106**, the source **116** can be formed as a second doped region **113** in the device layer **106**, and the gate **104** can be formed on the device layer **206**. The gate **104** can comprise a second material, such as but not limited to, a Group III-V material. Examples of Group III-V materials that may be used as the second material include, but are not limited to, Indium Phosphide (InP), Gallium Nitride (GaN), and gallium arsenide (GaAs). In an illustrative example, the second material is GaAs. A conductive channel **132** can be formed in a region in the device layer **106** based on applying a voltage bias to the gate **104**. In various examples, the conductive channel **132** can be a conductive path from the source **116** to the drain **124**. An optical waveguide **102** can be formed in the device layer **106** above the conductive channel **132**, for example, between the region in which the conductive channel **132** is formed and the gate **104**.
[0020] While the conductive channel **132** is schematically shown as an arrow, the actual shape of the conductive channel **132** is not intended to be limited to the depicted arrow. The conductive channel **132** may be dependent on the materials (and doping levels) of the optical device **100**, as well as the voltage bias applied across the device.
[0021] In an example, the first and second doped regions **123** and **113** comprise a first doping polarity. For example, the first material of the device layer **106** may be doped with the first doping polarity (e.g., p-type doping in one example) at a first doping concentration. The first and second doped regions **123** and **113**, corresponding to the drain **124** and source **116**, respectively, may be doped with the first doping polarity at a second and third doping concentration, respectively, each of which is higher than the first doping concentration (e.g., +p-type doping concentration in this example). In some examples, the second and third doping concentrations are approximately equal. In another example, the second and third doping concentrations need not be the same. The second material forming the gate **104** may comprise a second doping polarity that is the opposite of the first doping polarity (e.g., n-type doping in this example). In this case, when an appropriate voltage bias is applied to the gate **104**, as described below in greater detail, the conductive channel **132** (e.g., a p-type channel in the above examples) can be formed that permits current to flow from the source **116** to the drain **124** in a direction illustrated as arrow **134**. In another example, the doping polarities may be reversed, in which case the direction of the current may be reversed.
[0022] In some examples, a buried oxide (BOX) layer **101** may be grown on the underlying substrate **108** and the device layer **106** formed thereon. In an example, BOX layer **101** may comprise silicon dioxide (SiO.sub.2). Trenches **122** and **128**, provided for example as air gaps, separates the waveguide **102** from the first and second doped regions **123** and **113** forming the

drain **124** and the source **116**. The BOX layer **101** may be provided to confine an optical signal **105** propagating in the waveguide **102** in a vertical direction, while the trenches **122** and **128** may be provided to confine the optical signal **105** in a horizontal direction.

[0023] Optical device **100** may comprise one or more structures in addition to the MOSFET structure constituted as the source **116**, drain **124**, and gate **104**. For example, as shown in FIG. **1**A, optical device **100** may be a hybrid MOS device comprising a capacitive structure **110**. In this case, the capacitive structure **110** includes a cathode **112** comprising the second material and formed on the waveguide **102**. In the example of FIG. **1**A, the cathode **112** may be provided as the gate **104** or a portion of gate **104**. The optical device **100** also includes an anode **114** formed in the device layer **106**. The anode **114** comprises the waveguide **102**. That is, for example, the region of the device layer **106** provided as anode **114** comprises the waveguide **102**, as well as other portions of the device layer **106**. The anode **114** adjoins the cathode **112**, thereby defining a capacitive structure **110** between the anode **114** and the cathode **112**. In various examples, the cathode **112** comprises Group III-V material as the second material, in which case the capacitive structure **110** may be considered a MOSCAP.

[0024] A dielectric **120** is formed between the cathode **112** and the anode **114**. The dielectric **120** may be an electrically insulating material formed between the cathode **112** and anode **114** of the capacitive structure **110**, and the polarization of the dielectric **120** by an applied electric field may increase the surface charge of the capacitive structure **110** for a given electric field strength. The dielectric **120** can be native oxides of the cathode or the anode or both, or can be external dielectric materials such as high-k dielectrics or polymers which can be formed by deposition, oxidation, wafer bonding or other dielectric coating methods. In other examples, dielectric **120** may be formed from SiO.sub.2, Si.sub.3N.sub.4, Al.sub.2O.sub.3, polyimides, transition metal oxides (e.g., HfO.sub.2, titanium dioxide, zinc oxide, nickel oxide, etc.), organic materials, chalcogenides, 2D materials (e.g., molybdenum disulfide and the like), and ferroelectric materials, among others.

[0025] An electrode **126** may be disposed on the second doped region **113** forming the source **116** and an electrode **130** can be disposed on the first doped region **123** forming the drain **124**. When a voltage is applied, for example, by power source **136***a*, between the electrodes **126** and **130**, carrier accumulation or depletion can occur around dielectric **120**. Due to the capacitive structure **110** overlapping with the waveguide **102** (e.g., comprising the waveguide **102**), carrier concentration change may lead to changes in refractive index and propagation loss within waveguide **102**. By biasing the voltage applied between the electrodes **126** and **130**, the refractive index may be changed accordingly, thereby inducing a phase shift modulation. Thus, optical signal **105** propagating through waveguide **102** can be phase shifted based on changes in the refractive index induced by applying a voltage biasing to the capacitive structure **110**. The phase shifted optical signal **105** then continues along the waveguide **102**. As such, the optical device **100** can be used as an optical modulator.

[0026] For example, FIG. **1**A includes a power source **136***a*, which can be controlled by, for example, a control circuit **150**. The power source **136***a* can act as a signal source and has a negative terminal connected to the electrode **126** and a positive terminal connected the electrode **130**. This results in a migration of negative charges from the cathode **112** toward a side of the waveguide **102** adjacent to the cathode **112**, and migration of positive charges (“holes”) from the anode **114** to an opposite side of the waveguide **102** (also referred to herein as accumulation mode). In other examples the polarity of the power source **136***a* may be reversed. Reversing the polarity of the power source **136***a* causes a migration of negative charges from the waveguide **102** toward electrode **126**, and migration of holes from the waveguide **102** toward electrode **130** (also referred to herein as depletion mode).

[0027] The dielectric **120** forms at the boundary between the Group III-V material of the cathode **112** and the underlying capacitor portion of the device layer **106**. A thin layer (e.g., on the order of tens of nanometers or less) of silicon and the Group III-V oxides can be provided between the cathode **112** and anode **114**, from which dielectric **120** forms naturally at this boundary and serves as a dielectric for the capacitive structure **110**. In some examples, this thin layer has a thickness on a nanoscale, for example, a few nanometers thick. In some examples, steps need not be taken to encourage the formation of dielectric **120**. In other examples, the formation of dielectric **120** may be stimulated, for example by elevating the temperature, exposing the materials to an oxygen-rich atmosphere, or other suitable technique.

[0028] The capacitive structure **110** can operate in accumulation or depletion mode. As discussed above, a voltage bias can be applied between the anode and the cathode, causing a thin charge layer to accumulate or deplete on both sides of the dielectric **120**. The resulting change in free carrier density causes a change in refractive index (n) of the waveguide **102**, which is manifested as a change in the effective refractive index (Δn.sub.eff). The amount of change or modulation in the effective refractive index (Δn.sub.eff) and associated change in optical losses (Δα) can be described as follows:

[00001] $$n_{\text{eff}} = \frac{-q^{2}\ {}^{2}_{0}}{8\ {}^{2}c^{2}n\ {}_{0}}\left(\frac{N_e}{m^{*}_{ce}} + \frac{N_h}{m^{*}_{ch}}\right) \quad \text{Eq.1} \qquad \Delta\alpha = \frac{-q^{3}\ {}^{2}_{0}}{4\ {}^{2}c^{3}n\ {}_{0}}\left(\frac{N_e}{m^{*2}_{ce}\ {}_{e}} + \frac{N_h}{m^{*2}_{ch}\ {}_{h}}\right) \quad \text{Eq.2}$$

[0029] where q is electrical charge applied to the cathode **112** and the anode **114**; c is the speed of light in vacuum; ϵ.sub.0 is the permittivity of free space; n is the material refractive index; ΔN represents a change in carrier density, such that ΔN.sub.e represents the change in carrier density in terms of electrons and ΔN.sub.h represents the change in carrier density in terms of holes; m* represents the relative effective mass of electrons (m*.sub.ce) and holes (m*.sub.ch); μ.sub.h represents the hole mobility; μ.sub.e represents the electron mobility; and λ.sub.0 is the free space wavelength.

[0030] An optical phase shift (Δφ) in the waveguide **102** induced by the capacitive structure **110** depends on the magnitude of the voltage-induced Δn.sub.eff, the device length L, and the optical wavelength λ of the optical signal **105**. In this example, the optical phase shift can be calculated as Δφ=2π Δn.sub.effLλ. Thus, the optical phase of the light within waveguide **102** may be shifted based on the voltage-induced Δn.sub.eff for a given length L of the optical device **100**. Accordingly, the modulation depth of optical phase shift (Δφ) can be increased by increasing the length L of the optical device **100**, which can be detrimental in terms of physical footprint on a PIC and processing time.

[0031] Accordingly, optical device **100** can leverage the MOSFET formed thereon to increase the modulation depth of optical phase shift (Δφ) by increasing the voltage-induced Δneff of Eq. 1. As such, optical device **100** can be utilized at an optical modulator with improved optical modulation, For example, an electrode **138** may be disposed on the gate **104**. A gate voltage (V.sub.GS) can be applied, for example, by power source **136***b*, connected between the electrodes **138** and **126**. The gate voltage (V.sub.GS) can be applied in a manner to enter saturation mode and form the conductive channel **132**. By further modulating the gate voltage (V.sub.GS) within the saturation mode, a channel length of the conductive channel **132** can be modulated (e.g., through channel length modulation techniques as known in the art), which can function to alter carrier concentration within the region of the waveguide **102**. Increasing the carrier concentration within this region means increased density of carriers can be available for accumulation or depletion induced by the capacitive structure **110**. That is, for example, increasing the gate voltage (V.sub.GS) can increase carrier concentration, which alters the change in carrier density (ΔN) of Eq. 1 above and functions to increase the modulation depth.

[0032] For example, the MOSFET may comprise a threshold voltage (Vth) based on the materials and structural dimensions of the source **116**, drain **124**, and gate **104**. The power source **136***b*, which can be controlled by, for example, the control circuit **150** and can be used as a signal source having a negative terminal connected to the electrode **138** and a positive terminal connected the electrode **126**. The power source **136***a* can be controlled to tune the gate voltage (V.sub.GS). When the gate voltage (V.sub.GS) is greater than the threshold voltage (V.sub.th), but less than a drain to source voltage (V.sub.DS), the MOSFET operates in a linear region (known as triode or ohmic mode). During this mode of operation, current entering the drain increases linearly with drain to source voltage (V.sub.DS). When the gate voltage (V.sub.GS) is greater than the threshold voltage (V.sub.th) and equal to or greater than the drain to source voltage (V.sub.DS), the MOSFET operates in saturation mode. The onset of this mode is referred to as pinch-off, indicating a lack of a conductive channel near the drain. Upon switching to the saturation mode, the current in the drain remains constant with an increasing voltage applied across the drain and source, referred to as a drain-to-source voltage (V.sub.DS). Upon entering the saturation mode, the conductive channel **132** is formed having a channel length.

[0033] In some examples, the gate voltage (V.sub.GS) can be adjusted to further tune the channel length, for example, through channel length modulation, and carrier concentration in the region of the waveguide **102**. For example, in one case, the gate voltage (V.sub.GS) can be increased leading to increased channel length. In another example, gate voltage (V.sub.GS) can be decreased leading to a decrease in channel length. By decreasing the channel length, the number of carriers in the gate **138** can be reduced, which can lead to a larger extinction ratio and modulation depth in the modulator. Accordingly, optical device **100** can provide for channel length modulation as an alternative and/or additional type of modulation mechanism to carrier accumulation modulation that can be performed in the MOS capacitor.

[0034] In some examples the polarity of the power source **136***b* may be reversed, for example, when the first material is n-type doped. Reversing the polarity of the power source **136***b* causes the direction of the current to be reversed.

[0035] FIGS. **2**A and **2**B illustrate another example optical device **200** in accordance with implementations disclosed herein. Optical device **200** may be an example of a PIC that can be provided on a chip. In various examples, optical device **200** may be integrated into a silicon photonics platform.

[0036] Optical device **200** comprises a substrate **208** having a device layer **206** formed on a BOX layer **201**. The substrate **208**, device layer **206**, and BOX layer **201** may be substantially similar to substrate **108**, device layer **106**, and BOX layer **101** of FIG. **1**A, except as provided herein. For example, device layer **206** may comprise the first material.

[0037] The optical device **200** also comprises a MOSFET formed on substrate **208**. The MOSFET can be formed between a drain **224**, gate **204**, and source **216**, each of which may be substantially similar to drain **124**, gate **104**, and source **116** respectively. The drain **224** can be formed as a first doped region **223** in the device layer **206**, the source **116** can be formed as a second doped region **213** in the device layer **206**, and the gate **104** can be formed on the device layer **206**. The gate **204** can comprise a second material, such as but not limited to, a Group III-V material. A conductive channel **232** can be formed in a region in the device layer **206** based on applying a voltage bias to the gate **204**, for example, as described in connection with FIG. **1**A above. An optical waveguide **202** (e.g., substantially similar to waveguide **102** of FIGS. **1**A and **1**B) can be formed in the device layer **206** above the conductive channel **232**, for example, between the region in which the conductive channel **232** is formed and the gate **204**. The first and second doped reigns may be substantially similar to the first and second doped regions described above in connection with FIG. **1**A.

[0038] While the conductive channel **232** is schematically shown as an arrow, the actual shape of the conductive channel **232** is not intended to be limited to the depicted arrow. The conductive channel **232** may be dependent on the materials (and doping levels) of the optical device **200**, as well as the voltage bias applied across the device.

[0039] Trenches **222** and **228**, provided for example as air gaps, separate the waveguide **202** from the first and second doped regions **223** and **213**, respectively. The BOX layer **201** may be provided to confine an optical signal **205** propagating in the waveguide **202** in a vertical direction, while the trenches **222** and **228** may be provided to confine the optical signal **205** in a horizontal direction.

[0040] Optical device **200** may comprise one or more structures in addition to the MOSFET structure constituted as the source **216**, drain **224**, and gate **204**. For example, as shown in FIG. **2**A, optical device **200** may comprise a capacitive structure **210** that is substantially similar to the capacitive structure **110** of FIG. **1**A. For example, capacitive structure **210** includes a cathode **212** comprising the first material and formed on the waveguide **202**. The optical device **200** also includes an anode **214** formed in the device layer **206**. The anode **214** comprises the waveguide **202**. That is, for example, the region of the device layer **206** provided as anode **214** comprises the waveguide **202**, as well as other portions of the device layer **206**. The anode **214** and cathode **212** may be substantially similar to anode **114** and cathode **112**. For example, the anode **214** adjoins the cathode **212**, thereby defining the capacitive structure **210**, which may be implemented as a MOSCAP. Additionally, a dielectric **220** is formed between the cathode **112** and the anode **114**. The dielectric **220** can be substantially similar to dielectric **120** of FIG. **1**A.
[0041] Accordingly, the capacitive structure **210** can be utilized to induce a phase shift modulation in the waveguide **202**, for example, by applying a voltage bias across the first and electrodes **226** and **230**. That is, for example, power source **236***a* can be controlled by control circuit **250** to apply a voltage bias, which can be tuned to induce carrier accumulation or duplication around the dielectric **220**. Thus, in a manner similar to that described above in connection with FIG. **1**A, optical signal **205** propagating through waveguide **202** can be phase shifted based on changes in the refractive index induced by applying the voltage bias to the capacitive structure **210**.
[0042] Additionally, optical device **200** can leverage the MOSFET formed thereon to increase the modulation depth of optical phase shift by increasing the voltage-induced Δneff of Eq. 1. As such, the optical device **200** can be utilized at an optical modulator with improved optical modulation, That is, as explained above in connection with FIG. **1**A, an electrode **238** may be disposed on the gate **204** and a gate voltage (V.sub.GS) can be applied, for example, by power source **236***b* (e.g., substantially similar to power source **136***b*), connected between the electrodes **238** and **226**. The control circuit **250** can be used to control power source **236***b* so to apply a gate voltage (V.sub.GS) in a manner to operate the MOSFET in saturation mode and form the conductive channel **232**. The channel length of the conductive channel **232** can be modulated by increasing the voltage bias, which can function to alter carrier concentration within the region of the waveguide **102** and altering the change in carrier density (ΔN) of Eq. 1.
[0043] Additionally, in the example of FIG. **2**A, optical device **200** may comprise another capacitive structure **240** connected in series to the MOSFET. More particularly, as shown in FIG. **2**A, the capacitive structure **240** is connected in series to the drain **224** of the MOSFET. The capacitive structure **240** comprises a waveguide **203** formed in the device layer **206**. The capacitive structure **240** can include a cathode **209** comprising the second material and formed on the waveguide **203**. In the example of FIG. **2**A, electrode **230** is formed on the cathode **209**. The capacitive structure **240** also includes an anode **217** formed in the device layer **206**. The anode **217** comprises the waveguide **203**. That is, for example, the region of the device layer **206** provided as anode **217** comprises the waveguide **203**, as well as other portions of the device layer **206**. The anode **217** may be connected to the electrode **226** via the device layer **206** and connected in series with the drain **224**, for example, via the cathode **209**. The anode **217** adjoins the cathode **209**, thereby defining the capacitive structure **240** between the anode **217** and the cathode **209**. In various examples, the cathode **209** comprises Group III-V material as the second material, in which case the capacitive structure **240** may be considered a MOSCAP.
[0044] As with capacitive structure **210**, the dielectric **220** is formed between the cathode **209** and the anode **217**. The dielectric **220** may be an electrically insulating material formed between the cathode **209** and the anode **217**, and the polarization of the dielectric **220** by an applied electric field may increase the surface charge of the capacitive structure **240** for a given electric field strength. In various examples, the cathode **209** and gate **204** are formed of the same material, in which case the dielectric **220** may be a connected layer formed of a common oxide. In another

example, the cathode **209** and gate **204** may be different Group III-V materials, in which case it may be beneficial to provide the dielectric **220** as separate dielectrics having oxides for the materials forming the cathode **209** and gate **204**.

[0045] In some examples, the capacitive structure **240** may be leveraged as a memristor (in which case the capacitive structure **240** may be referred to as a memristive structure). In this case, the waveguide **203** is formed of a third doped region **225** above the channel region. The third doped region **225** may comprise the first doping polarity and doped with a fourth doping concentration. In some examples, the fourth doping concentration may be approximately the same as the third doping concentration or second doping concentration. Thus, in an illustrative example, the third doped region **225** may comprise p-type doping having a concentration higher than the first doping concentration of the device layer **206**.

[0046] Applying an electric field across the capacitive structure **240** can form a filamentation layer in the capacitive structure **240** that provides for non-volatile memristive behavior. For example, power source **236***c* can be connect to electrode **230** and an optional electrode **249**. The power source can be controlled by control circuit **250** to apply a voltage bias across the capacitive structure **240**, which generates charge trap regions **247** forming conductive path **245** within the BOX layer **201**. The capacitive structure **240** may be considered to be in an “on” state at this point providing memristive behavior (e.g., non-volatile retention). The conductive path **245** exhibits conductive behavior within the capacitive structure **240**. The conductive path **245** and charge trap regions **247** remain even after the voltage bias supplied by power source **236***c* is removed, thereby providing non-volatile retention.

[0047] The conductive path **245** formed in the capacitive structure **240** causes the effective refractive index of the waveguide **203** to change. Thus, selective control of the voltage bias applied to capacitive structure **240** through power source **236***c* can be used to form a desired conductive path **245**, which tunes the effective refractive index of waveguide **203** and sets an adjustment of the phase of an optical signal on waveguide **203** induced by the changed effective refractive index. Since the conductive path **245** remains when voltage bias is not applied to the capacitive structure **240**, tuning of the capacitive structure **240** can be achieved through non-volatile memristive behavior and maintained without a connected power source.

[0048] Resetting of the induced phase shift can be reversed by applying a voltage bias by power source **236***c* having a reverse polarity relative to the voltage bias used to form the conductive path **245**. The reverse voltage bias can rupture the conductive path **245** by dissipating the charge trap regions **247** and permitting a charge to relocate within the capacitive structure **240**. This effectively restores the capacitive structure **240** to the “off” state.

[0049] In more detail, electroforming or filamentation can be stimulated by applying an electric field (referred to herein as a “set electric field” or “set voltage bias”) across BOX layer **201**. The magnitude of the applied electric field may be large enough to stimulate electroforming. For example, by increasing the voltage bias supplied by power source **236***c* to the optional electrode **249** and electrode **230**, electroforming can be stimulated in the BOX layer **201** which forms filaments in the dielectric **220** that extend between cathode **209** and anode **217**, shown in zoomed in view of capacitive structure **240** as filaments **243**. The filaments **243** are depicted as having a shape, but this is for illustrative purposes only and not intended to illustrate the actual shape or structure of filaments **243**. The specific shape and structure of the filaments **243** may be dependent on materials (and doping of the materials) forming the various layers and the voltage bias applied to the electrodes **249** and **226** that form the filaments **243**.

[0050] When the filaments **243** are present, the dielectric **220** may be considered a filamentation layer. The filamentation layer provides an electrical path between cathode **209** and anode **217** that causes charge trap regions **247** to form in the BOX layer **201**, which creates a conductive path **245** (shown as an arrow) within the underlying BOX layer **201**. In one example, the filamentation layer may create defect sites within the BOX layer **201** that function to trap charges. The charge trap regions **247** operate to hold negative charges within the conductive path **245**, thereby inducing a migration of negative charges. This resulting change in free carrier density causes the change in the refractive index of waveguide **203**, similar to that described above in connection with Eqs. 1 and 2. Thus, the accumulation of charges BOX layer **201** induces an amount of change in the effective refractive index, which can be selectively set to induce a desired phase shift in an optical signal **207** propagating in waveguide **203**, as described above.

[0051] While the conductive path **245** is schematically shown as an arrow, the actual shape of the conductive path **245** is not intended to be limited to the depicted arrow. The conductive path **245** may be dependent on the materials (and doping levels) of the optical device **200**, as well as the voltage bias applied across the device. Similarly, conductive channel **232** is not limited to the depicted arrow and may be dependent on the materials (and doping levels) of the optical device **200**, as well as the voltage bias applied across the device.

[0052] In examples, the conductive path can effectively change the resistance of the BOX layer **201**, which causes the capacitive structure **240** to switch from capacitive operation to a resistive operation (e.g., the capacitive structure effectively functions as resistor). This change in operation exhibits a conductive behavior provided by filaments **243** subject to a resistance provided by the dielectric layer **220**. The conductive path **245** and charge trap regions **247** can remain in the underlying BOX layer **201** in the absence of a voltage bias, thereby providing non-volatile retention. The conductive path **245** causes a further change in the effective index of refraction (n.sub.eff) of the waveguide **203**, which can be used to tune the phase shift (Δφ) induced by the optical device **200**. Since the conductive path **245** remains when voltage bias is not applied, phase tuning of optical signal **207** in waveguide **203** can be achieved through non-volatile memristive behavior and maintained in the absence of power supplied to the optical device **200**.

[0053] As shown in the zoomed in region of capacitive structure **240**, a plurality of layers are provided, for example, at the interface between the material of the anode **217** and the material of the cathode **209**. The dielectric **220** can be formed at this interface between cathode **209** and anode **217**. The interface comprises a plurality of layers **220***a*-**220***d* and a bonding interface **221** resulting from wafer bonding of the anode **217** to cathode **209**. In various examples, the cathode **209** may be hydrophilic wafer-bonded to the anode **217**, such that the material of cathode **209** is bonded to the waveguide **203** resulting in bonding interface **221**.

[0054] Prior to bonding, layers **220***a*-**220***d* are provided that, once bonded, stimulate formation of the dielectric **220**, as discussed above. For example, layer **220***d* can be deposited on anode **217** followed by layer **220***b*. Similarly, layer **220***c* can be deposited on cathode **209** followed by layer **220***a*. The materials of layers **220***a*-**220***d* may include, but not limited to, SiO.sub.2, Si.sub.3N.sub.4, Al.sub.2O.sub.3, HfO.sub.2, polyimide, BCB, polyimides, transition metal oxides (e.g., titanium dioxide, zinc oxide, nickel oxide, etc.), organic materials, chalcogenides, 2D materials (e.g., molybdenum disulfide and the like), and ferroelectric materials, and combinations thereof. In some examples, layers **220***b* and **220***a* may be the same material, which may be different than the materials of layers **220***c* and **220***d*. In an example implementation, anode **217** may be coated with a layer of SiO.sub.2 as layer **220***d* followed by a layer of HfO.sub.2 as layer **220***b*, while cathode **209** may be coated with a layer of Al.sub.2O.sub.2 as layer **220***c* and followed by a layer of HfO.sub.2 as layer **220***a*. Each layer may be formed via atomic layer deposition (ALD) or other fabrication techniques as known in the art.

[0055] As noted above, the dielectric **220** may be formed between the cathode **209** and the anode **217**. The layers **220***a-d* can act to stimulate the formation of the dielectric **220**. As discussed above, dielectric **220** may be a thin layer of silicon and Group III-V oxides that forms naturally at the boundary and serves as dielectric **220**. In this example, the thin layer is provided as Al.sub.2O.sub.3 and HfO.sub.2, at the boundary of the anode **217** (e.g., between anode **217** and layer **220***b*), and the dielectric **220** is formed from these layers. In some examples, steps need not be taken to encourage the formation of dielectric **220**. In other examples, the formation of dielectric

220 may be stimulated, for example by elevating the temperature, exposing the materials to an oxygen-rich atmosphere, or other suitable technique.

[0056] When a high enough voltage bias is supplied across BOX layer **201** to stimulate electroforming, filaments **243** may be formed that transition dielectric **220** to a filamentation layer. As discussed above, formation of the filamentation layer results in formation of the conductive path **245** within underlying BOX layer **201**.

[0057] Furthermore, the non-volatile phase shift can be reset by applying a voltage bias (e.g., a reset voltage bias) having a reverse polarity relative to the set voltage bias. This reset voltage bias ruptures the filaments **243**, thereby removing the filamentation layer. As a result, the conductive path **245** in the BOX layer **201** may be ruptured by dissipating the charge trap regions **247** and permitting charge to relocate within the BOX layer **201**. This effectively restores the capacitive structure **240** to a low conductivity state, at which the capacitive structure **240** acts as a capacitor.

[0058] As described above, a larger voltage biases applied to a memristor, such as the capacitive structure **240** implemented as a memristive structure described above, can induce current flow beyond a current limit. This current limit may be dependent on the materials (and doping levels) of the optical device **200**. Exceeding the current limit can cause thermal break down, which can render the device permanently unusable. Accordingly, optical device **200** can utilize the MOSFET formed therein to induce current compliance by constraining the current in the capacitive structure **240** to magnitudes blow the current limit, thereby mitigating thermal break down in order to protect the device.

[0059] For example, the MOSFET can be operated to limit the current flowing through the capacitive structure **240** by applying a gate voltage (V.sub.GS) to gate **204** to transition the MOSFET into saturation mode. That is, for example, power source **236***b* can be controlled to apply the gate voltage (V.sub.GS) to gate **204**. At the onset of the saturation mode, pinch off occurs that limits the amount of current that can flow into the capacitive structure **240** and creating the conductive channel **232**, as described above. Carriers can then flow from source **216** to drain **224** and into to the capacitive structure **240**. When the filaments **243** are formed (e.g., the capacitive structure **240** is providing non-volatile retention) in a low resistance state, current flow passes through the waveguide **203**, which changes the overall effective refractive index of the waveguide **203** in accordance with Eq. 1 and 2 described above. Using the MOSFET, the current can be controlled by controlling the gate voltage (V.sub.GS), and subsequently, controlling the resistance state of the memristor, along with the amount of effective refractive index change in the waveguide **203**.

[0060] Accordingly, in various examples, the MOSFET of optical device **200** can be operated (e.g., by applying a gate voltage) to limit an amount of current flowing through the memristor, so it can control the resistance of the memristor. This can occur by applying a set voltage bias on the memristor, which causes the current to reach the saturation current of the MOSFET (e.g., saturation mode) and setting the resistance of the device higher or lower depending on the saturation current value. For example, the MOSFET can be controlled for current compliance (e.g., limiting current flow to less than the current limit), which can be used to control the switching of the resistive state of the memristor (e.g., a higher current compliance can lead to a lower resistance state, and a lower current compliance will lead to a higher resistance state). This can occur by forming the filamentation layer with the current compliance.

[0061] Optical device **200** may also include trenches **235** and **237**. Trenches **235** and **237** may be provided for example as air gaps that can operate to confine the optical signal **207** in a horizontal direction. Similarly, the BOX layer **201** confine the optical signal **207** in a vertical direction.

[0062] According to some implementations, one or more of the capacitive structures of the example optical device **200** can be utilized for detecting optical signals without extracting light out of a coupled waveguide. For example, FIG. 2B depicts an example implementation of optical device **200** in which the capacitive structure **210** can be implemented to detect optical power of, for example, the optical signal **205** propagating in waveguide **202**, without extracting optical signal **205** from the waveguide **202**. In this example, the capacitive structure **210** can function as a photodetector based on mid-bandgap defects that convert absorbed photons (e.g., from a received optical signal) into photo-generated carriers. Note that the power source **236***c* and reference numbers of the capacitive structure **240** have been removed in FIG. **2**B to assist with ease of illustration and clarity in the depicted figure. Furthermore, FIG. **2**B illustrates an example implementation in which the optional electrode **249** is removed.

[0063] The waveguide **202** of capacitive structure **210** may have one or more defect sites **233** as depicted in an enlarged view of waveguide **202**. The term "defect sites" as used herein may refer to imperfections in the bulk of the material of the waveguide **202**, surface imperfections at the boundaries of the waveguide **202**, or both. In some examples, the defect sites **233** may be resulted from imperfections in the manufacturing process as well as intentionally created. The existence of the defect sites **233** may absorb photons and cause the generation of free charge carriers relative to the intensity of the optical signal impinging thereon inside the waveguide **202**. A conductance of the waveguide **202** depends on the amount of the free charge carriers, such that an increase in the optical signal causes an increase in the conductance of the waveguide **202**. Accordingly, upon application of voltage bias across the optical device **200**, the defect sites **233** can cause the generation of free charge carriers relative to the intensity of the optical signal inside the waveguide **202**. The changes in the conductance within waveguide **202** may cause variations in current passing through the waveguide **202** which can be utilized to monitor the intensity of the optical signal **205** propagating in waveguide **202**. For example, a decrease in current passing through the capacitive structure **210** is a result of proportional decrease in conductance because the intensity of the optical signal **205** may have also decreased, which means fewer free charge carriers are generated to change the conductance. This change can be monitored to detect an optical power of the optical signal **205**.

[0064] In some examples, as shown in FIG. **2**B, the optical device **200** can be connected to control circuit **250**, which can be used as a control circuit configured to monitor intensity of optical signals propagating within the waveguide **202** based on the detecting variations in current passing through capacitive structure **210**. The control circuit **250** may be electrically coupled to the capacitive structure **210**. For example, control circuit **250** may be coupled using electrical connection to electrodes **226** and **238**, which may represent monitoring sites either collectively or individually. In some examples, control circuit **250** may also be implemented as a control circuit for controlling one or more of power sources **236***a* and **236***b*, as described above.

[0065] In some examples, the control circuit **250** may cause capacitive structure **210** to generate electrical signals indicative the intensity of the optical signal propagating within waveguide **202**. To affect the generation of the electrical signals, in some examples, the control circuit **250** may include a lock-in amplifier **254** and a pre-amplifier **252**. In some examples, the lock-in amplifier **254** may generate a reference variable voltage signal, for example, a sinusoidal signal. The lock-in amplifier **254** may determine a frequency of the reference variable voltage signal based on a conductance of waveguide **202** in the given capacitive structure **210** and a capacitance of capacitive structure **210**. In an example, the lock-in amplifier **254** may determine a frequency (F.sub.0) of the reference variable voltage signal based on an example relationship:

[00002] $F_0 = \frac{G_{WG}}{2 * C} \quad \mathrm{Eq.}\,3$

[0066] where, G.sub.WG represents the conductance of the waveguide region in the capacitive structure **210** and C represents the capacitance of the capacitive structure **210**. In certain other examples, the frequency (F.sub.0) of the reference variable voltage signal may be set to any value greater than.

[00003] $\frac{G_{WG}}{2 * C}.$

[0067] The control circuit **250** may apply a reference variable voltage signal to capacitive structure

210 via electrode **226** and **238**. The conductance of the waveguide **202** may change depending on the intensities of the optical signal therein. Consequently, the current flowing through the waveguide **202** within may also vary. In particular, the magnitude of electrical current generated by capacitive structure **210** may be influenced by the light intensities within the waveguide **202**, because the waveguide conductance changes due to the presence of free carriers created by the absorption of photons at the defect sites **233**.

[0068] The control circuit **250** may receive the electrical signal generated within a capacitive structure **210** and measure the electrical signals (e.g., electrical currents). The electrical signals are in turn representative of intensities of the optical signals inside a waveguide **202** formed in the capacitive structure **210**. In some examples, the electrical currents received by the control circuit **250** from the capacitive structure **210** may be weak in strength. The pre-amplifier **252** may amplify the electrical currents for further processing by the lock-in amplifier **254**.

[0069] During operation, to monitor the optical power (e.g., light intensity) of optical signal **205**, the control circuit **250** may apply the reference variable voltage signal (e.g., sometimes referred to herein as a reference bias) to electrodes **226** and **238** and measure electrical current at the one of the electrodes. As previously noted, changes in the signal intensities inside the waveguide **202** may cause changes in the conductance of the waveguide **202**. Consequently, the current flowing through the respective capacitive structure **210** may vary. Such current flowing through the respective capacitive structure **210** may be monitored by the control circuit **250** at the one of the electrodes.

[0070] While FIG. **2**B depicts capacitive structure **210** leveraged to monitor optical power of optical signal **205** in waveguide **202**, examples disclosed herein are not limited to only capacitive structure **210**. In some examples, capacitive structure **240** can be used to monitor the optical power of optical signal **207** in waveguide **203**, for example, based on defect sites contained in the waveguide **203**. In some examples, both capacitive structures **210** and **240** may be used to monitor optical power of an optical signal in a respective waveguide **202** and **203**.

[0071] FIG. **3** depicts a schematic diagram of an optical device **300** in implemented as a Mach-Zehnder Interferometer in accordance with an example disclosed herein. Optical device **300** may be an example PIC that can be provided on a chip. In various examples, optical device **300** may be integrated into a silicon photonics platform.

[0072] The optical device **300** according to the example shown in FIG. **3** comprises a MZI **310** and the optical device **200**, depicted as a top down view of the optical device **200**. Arrow **234** represents a direction of current flow according to the examples described above in connection with FIG. **2**A, for example, the direction of current flow of conductive channel **232**. FIG. **3** also illustrates line B-B′, which can be an example line along with the cross section of optical device **200** is taken.

[0073] Optical device **300**, according to this example, is provided as MZI **310** comprising waveguides **303** and **302** that guide propagation of an optical signal input into an input port **312***a* and **312***b*, respectively. For example, MZI **310** may comprise a arm **311** formed of a waveguide **303** having input port **312***a* and arm **313** formed of a waveguide **302** having input port **312***b*.

[0074] An optical signal input, for example by an optical source (not shown) into an input port of one of the waveguides can be optically coupled into the other waveguide by a coupler **316**. An amount of the optical signal coupled into one waveguide from the other is based on a coupling efficiency of coupler **316**. For example, a portion of an optical signal supplied to input port **312***a* can be coupled into waveguide **302** according to the coupling efficiency of the coupler **316**. In various examples, the coupling efficiency of coupler **316** may be 50%, such that equal amount of the input optical remains in waveguide **303** (e.g., in arm **311**) and is coupled into waveguide **302** (e.g., into arm **313**). Coupler **316**, according to illustrative examples, may be a directional coupler configured to evanescently couple an optical signal between adjacent waveguides.

[0075] In the example of FIG. **3**, the optical device **200** is integrated with the MZI **310**, such that the waveguides **302** and **303** are coupled to the optical device **200**. In the example of FIG. **3**, the power sources and corresponding connections are not shown for illustrative purposed only. In this case, the waveguide **303** may be an example of waveguide **203** of FIG. **2**A and the waveguide **302** may be an example of waveguide **202** of FIG. **2**A. Thus, an optical signal propagating in the arm **311** may be an example of optical signal **207**, while an optical signal propagating in the arm **313** may be an example of optical signal **205**.

[0076] As such, the optical device **200** can be utilized as described above in connection with FIG. **2**A. For example, a voltage bias can be applied between electrodes **230** and **226** to induce a change in the effective index of refraction of waveguide **302**, which alters optical phase shift (Δφ) in the optical signal therein, as described above in connection with FIGS. **1**A-**2**A. Furthermore, a voltage bias can be applied across electrodes **226** and **238** to form the conductive channel **232** in the MOSFET of optical device **200**, thereby increasing the modulation depth of the optical phase shift (Δφ) in the waveguide **302**, as described above in connection with FIGS. **1**A-**2**A.

[0077] The optical phase shift provided in waveguide **302** can be utilized to encode/decode information on optical signals based on destructive/constructive interface at the output of the MZI **310**. For example, information can be encoded/decoded by modulating the phase of waveguide **302** relative to the waveguide **303**. The optical signals on each arm **311** and **313** can be combined by a coupler **318**, which may be substantively similar to coupler **316**, resulting in constructive and/or destructive interference between optical signals from each arm. The interference causes amplitude modulation, which can encode information onto and/or decode information from the optical signal input in the MZI **310**, for example, by monitoring optical power at an output port **314***a* and/or **314***b*.

[0078] In some examples, in combination or alternatively with the above example, the optical device can comprise non-volatile retention as described above in connection with FIG. **2**A. For example, a voltage can be applied across the capacitive structure **240** (not highlighted in FIG. **3** as the various layers making up capacitive structure **240** include and are under the cathode **209**) via optional electrode **249** and electrode **230**, to operate the device in low resistance state and provide non-volatile memristive behavior with the capacitive structure **240**. The capacitive structure **240**, in this example, comprises the waveguide **303** as an example implementation of waveguide **203**.

[0079] In one example, the memristive behavior may provide for non-volatile phase tuning, which can provide for passband tuning for optical filtering by the optical device **200**. For example, MZI **310** may receive an input optical signal at the input port **312***a*, which is split into arm **311** and arm **313** according to a coupling coefficient of coupler **316**. The portion of waveguide **303** overlapping with the optical device **200** may experience a phase shift due to the memristive behavior, as described above in connection with FIG. **2**A. The phase shift can be controlled based on adjusting the voltage bias across the optional electrode **249** and electrode **230** to tune a resonance frequency via a setting voltage bias, thereby inducing a non-volatile phase-shift applied to the optical signal propagating along arm **311**. The optical signals propagating in the arms **311** and **313** can then be combined by a coupler **318**, resulting in constructive and/or destructive interference between optical signals from each arm.

[0080] In some example use cases, non-volatile phase tuning provided the memristive behavior can be used in photonic neural networks. For example, non-volatile phase tuning of a given optical device can be implemented to store a weight of the neural network. A collection of optical devices according to this example can store a number of weights enabling in-memory photonic processing.

[0081] Additional examples of memristive behavior can be found in, for example, U.S. application Ser. No. 18/382,657, the disclosure of which is incorporated herein by reference in its entirety.

[0082] In some examples, optical device **200** can be utilized for monitoring optical signals in the MZI **310** without extracting light out of a waveguides **302** or **303**. For example, as described above in connection with FIG. **2**B, the capacitive structure **210** can function as a photodetector based on mid-bandgap defects that convert absorbed photons (e.g., from a received optical signal) into photo-generated carriers, for example, based on defect sites comprised in a waveguide. In an example implementation, waveguide **302** may contain defect sites (e.g., defect sites **233**) that can cause the generation of free charge carriers relative to the intensity of the optical signal inside the waveguide **302**. Changes in the conductance within waveguide **302** may cause variations in current passing through the waveguide **302** which can be utilized to monitor the intensity of an optical signal propagating in waveguide **302**, for example, using a feedback or control circuit (e.g., control circuit **250**) as described above in connection with FIG. **2**B.

[0083] While the foregoing describes monitoring optical power of optical signal in waveguide **302**, examples disclosed herein are not limited thereto. In some examples, the optical power on waveguide **302** or a combination of waveguides **302** and **303** may be monitored, for example, based on defect sites contained in waveguides **303** and/or **304**.

[0084] By monitoring the optical power in one waveguide, examples disclosed herein can also monitor the optical power of an optical signal on the other waveguide. For example, as described above, coupler **316** can couple a portion of an optical signal supplied to input port **312***a* into waveguide **302** and the remainder of the optical signal continues along waveguide **303**. Thus, the optical power on waveguide **303** can be estimated based on detecting the optical power on the waveguide **302**. Estimating the optical power, in this example, on waveguide **303** is based on the optical power detected in waveguide **302** and the coupling efficiency. For example, in a case where the coupling efficiency of coupler **316** may be 50%, equal amounts of the input optical in waveguide **303** (e.g., in arm **311**) and in waveguides **302** (e.g., into arm **313**). In this case, detecting an optical power in arm **313** informs on the optical power the in arm **311**, which may be expected to be approximately equal.

[0085] In some examples, monitoring the optical power in a waveguide can be used to monitor and tune optical phase shift induced by the waveguide. For example, as described, a voltage bias can be applied between electrodes **230** and **226** to induce a change in the effective index of refraction of waveguide **302**, which alters optical phase shift ($\Delta\varphi$). By monitoring the optical power in the waveguide **302** as described above, the magnitude of voltage bias can be adjusted to a desired amount of optical phase shift. Similarly, in the case of setting and/or resetting non-volatile retention of the memristive structure, optical power monitored in waveguide **302** and be used to inform the whether the voltage bias applied across electrodes **249** and **230** is sufficient set or reset the device.

[0086] FIG. **4** illustrates an example optical device **400** implemented as a ring resonator in accordance with an example disclosed herein. Optical device **400** may be an example PIC that can be provided on a chip. In various examples, optical device **400** may be integrated into a silicon photonics platform.

[0087] The optical device **400** is an example of a resonator-based implementation of the optical device **200** of FIGS. **2**A and **2**B. In some examples, optical device **400** is implemented as a microring resonator (MRR). Optical device **400** includes a resonator structure **403** optically coupled to the bus waveguide **402**. In the example of FIG. **4**, the resonator structure **403** is provided as a resonator structure **403** that evanescently couples to the bus waveguide **402**. The bus waveguide **402** may be an example implementation of waveguide **202** of FIGS. **2**A and **2**B, and resonator structure **403** may be an example implementation of waveguide **203**. Thus, for example, bus waveguide **402** and resonator structure **403** may be formed in a device region, such as device layer **206** of FIG. **2**. The device layer or device region is not shown in FIG. **4** for illustrative purposes only so to clearly depict the various structures formed in the device layer. However, it will be appreciated that the device layer (or device region as referred to herein) would be present in forming the optical device **400** similar to optical device **100** and/or **200** described above.

[0088] The resonator structure **403** may be a closed loop structure around a central axis **490** having a shape that is circular, thereby forming a ring resonator or cavity. However other shapes are possible, for example but not limited to, elliptical, a racetrack shape, etc. An optical signal (e.g., optical signal **205**) may be received at an input end of the bus waveguide **402** and a portion coupled into resonator structure **403** (e.g., optical signal **207** in this example) based on a coupling efficiency of the evanescent coupling. The optical signal propagating in resonator structure **403** resonates therein and is coupled back onto the bus waveguide **402**, thereby combining the optical signal from resonator structure **403** with that on bus waveguide **402**. The combined optical signal can be output from the output end of the bus waveguide **402**. In various implementations, the resonator structure **403** and bus waveguide **402** may be formed of a semiconductor material, such as silicon or other Group IV material.

[0089] The resonator structure **403** may have a resonance wavelength defined based on the round-trip length of the resonator structure. In the case of evanescent coupling, the amount of light optically coupled into and out of the resonator structure **403** is based on a distance between the waveguides and the resonance frequency of the resonator structure **403**. The amount of light optically coupled can be represented by a coupling coefficient. The distance between waveguides can be controlled so to bring the resonator structure **403** as close as possible to bus waveguide **402**. The coupling length represents an effective curve length of the resonator structure **403** for the coupling phenomenon to happen with the waveguide, which can be based on a radius of curvature of the resonator structure **403** where the evanescent coupling is to occur. In the case of FIG. **4**, the radius of the resonator structure **403** may be designed to achieve a desired curve length for a desired resonance frequency. The resonance frequency is also dependent on effective refractive indices between the bus waveguide **402** and the resonator structure **403**.

[0090] In the example disclosed herein, the optical device **400** includes a MOSFET formed on a substrate (not shown in FIG. **4** for illustrative purposes only). The substrate may be substantially similar to the substrate **108** and/or **208** of FIGS. **1-2**B. The MOSFET can be formed between a source **416***a*, gate **404***a*, and drain **424**. The source **416***a*, gate **404***a*, and drain **424** may be example implementations of source **216**, gate, **204**, and drain **224**, respectively. Thus, for example, the drain **424** can be formed as a first doped region in a device region, the source **416***a* can be formed as a second doped region in the device region, and the gate **404***a* can be formed on the device region. The gate **404***a* can comprise the second material, such as but not limited to, a Group III-V material. A conductive channel **432** (shown as an arrow) can be formed in the device region between the source **416***a* to the drain **424** based on applying a voltage bias to the gate **404***a*. The bus waveguide **402** can be formed between the region in which the conductive channel **232** is formed and the gate **404***a*.

[0091] While the conductive channel **432** is schematically shown as an arrow, the actual shape of the conductive channel **432** is not intended to be limited to the depicted arrow. The conductive channel **432** may be dependent on the materials (and doping levels) of the optical device **400**, as well as the voltage bias applied across the device.

[0092] In the example of FIG. **4**, optical device **400** comprises additional structures in addition to the MOSFET structure constituted as the source **416***a*, drain **424**, and gate **404***a*. For example, optical device **400** comprises a capacitive structure **410***a*, which is substantially similar to the capacitive structure **210** of FIGS. **2**A and **2**B, and comprises the gate **404***a*. For example, capacitive structure **410** includes the gate **404***a* acting as a cathode formed on the bus waveguide **402**. The optical device **400** also includes an anode formed in the device region and comprises the bus waveguide **402**. The anode and cathode may be substantially similar to anode **214** and cathode **212** of FIGS. **2**A and **2**B. For example, the anode adjoins the cathode, thereby defining the capacitive structure **410**, which may be implemented as a MOSCAP. Additionally, a dielectric **420***a* is formed between the cathode (e.g., the gate **404***a*) and the anode (e.g., the bus waveguide **402**). The dielectric **420***a* can be substantially similar to dielectric **220** of FIGS. **2**A and **2**B.

[0093] Accordingly, the capacitive structure **410** can be utilized to induce a phase shift modulation in the bus waveguide **402**, for example, by applying a voltage bias across the first and electrodes **426***a* and **430**. That is, for example, a power source (not shown in FIG. **4**, but substantially similar to power source **236***a* of FIGS. **2**A and **2**B) can connected to terminals **482***a* and **482***b* and controlled (e.g., by a control circuit) to apply a voltage bias, which can be tuned to induce carrier accumulation or duplication around the dielectric **420***a*. Thus, as described above, an optical signal on bus waveguide **402** can be phase shifted based on changes in the refractive index induced by applying the voltage bias to the capacitive structure **410**.

[0094] The MOSFET of optical device **400** can be driven to increase the modulation depth of optical phase shift by increasing the voltage-induced Δneff of Eq. 1. That is, as explained above in connection with FIGS. **1**-**2**B, an electrode **438***a* having terminal **482***c* may be disposed on the gate **404***a* and a gate voltage (V.sub.GS) can be applied, for example, by a power source (e.g., not shown but substantially similar to power source **236***b*), connected between the electrodes **438***a* and **426***a*. The power source may be controlled (e.g., by a control circuit) so to apply a gate voltage (V.sub.GS) in a manner to operate the MOSFET in saturation mode and form the conductive channel **432**. The channel length of the conductive channel **432** can be modulated by increasing the voltage bias, which can function to alter carrier concentration within the region of the bus waveguide **402** and altering the change in carrier density (ΔN) of Eq. 1.

[0095] Additionally, in the example of FIG. **4**, optical device **400** may comprise another capacitive structure **440** connected in series to the MOSFET. More particularly, the capacitive structure **440** can be connected in series to the drain **424** of the MOSFET. The capacitive structure **440** comprises the resonator structure **403** and a cathode **409** formed on the waveguide **403**. The cathode **409** may comprise the second material and be substantially similar to the cathode **209** of FIGS. **2**A and **2**B. In the example of FIG. **4**, electrode **430** is formed on the cathode **409**, for example, within a central region of the resonator structure **403** (e.g., along the central axis **490** as shown in the example of FIG. **4**). The capacitive structure **440** also includes an anode formed in the device region and comprises the resonator structure **403**. The anode (e.g., resonator structure **403**) may be connected to the electrode **426***a* via the device region and connected in series with the drain **424**, for example, via the cathode **409**. The anode (e.g., resonator structure **403**) adjoins the cathode **409**, thereby defining the capacitive structure **440** between the anode and the cathode **409**. In various examples, the cathode **409** comprises Group III-V material as the second material, in which case the capacitive structure **440** may be considered a MOSCAP.

[0096] As with capacitive structure **410**, a dielectric **420***c* is formed between the cathode **409** and the anode (e.g., resonator structure **403**). The dielectric **420***c* may be an electrically insulating material formed between the cathode **209** and the anode (e.g., resonator structure **403**), and the polarization of the dielectric **420***c* by an applied electric field may increase the surface charge of the capacitive structure **440** for a given electric field strength. In various examples, the cathode **409** and gate **404***a* are formed of the same material, in which case the dielectric **420***c* may be a connected layer formed of a common oxide. In another example, the cathode **409** and gate **404***a* may be different Group III-V materials, in which case it may be beneficial to provide the dielectric **420***c* as separate dielectrics having oxides for the materials forming the cathode **409** and gate **404***a*.

[0097] In the example of FIG. **4**, the dielectric **420***c* and cathode **409** are shown with a portion remove for illustrative purposes. That is, for example, the dielectric **420***c* and cathode **409** may be provided as cylindrical structures having a full circular profile when viewed along central axis **490**. In order to show the resonator structure **403** formed under the dielectric **420***c*, a piece of the dielectric **420***c* and cathode **409** is not shown.

[0098] The optical device **400**, according to some example, can be utilized at an optical modulator with improved optical modulation, described above in connection with FIG. **2**A. For example, a voltage bias can be applied between electrodes **430** and **426***a* to induce a change in the effective index of refraction of waveguide **402**, which alters optical phase shift (Δφ) in the optical signal therein, as described above in connection with FIGS. **1**A-**2**A. Furthermore, a voltage bias can be applied across electrodes **426***a* and **438** to form the conductive channel **432** in the MOSFET of optical device **400**, thereby increasing the modulation depth of the optical phase shift (Δφ) in the waveguide **402**, as described above in connection with FIGS. **1**A-**2**A.

[0099] The optical phase shift provided in waveguide **402** can be utilized to encode/decode information on optical signals based on destructive/constructive interface at the output of the optical device **400**. For example, information can be encoded/decoded by modulating the phase of waveguide **402** relative to the waveguide **403**, resulting in constructive and/or destructive interference between optical signals. The interference causes amplitude modulation, which can encode information onto and/or decode information from the optical signal input in to waveguide **402**.

[0100] In some examples, the capacitive structure **440** may be leveraged as a memristor to provide non-volatile retention. In this case, the resonator structure **403** may be formed of a third doped region, as described above in connection with FIG. **2**A. Applying an electric field across the capacitive structure **440** can form a filamentation layer in the capacitive structure **440** that provides for memristive behavior in the form of non-volatile retention. For example, as described above, a power source (not shown in FIG. **4** but substantially similar to power source **236***c*) can be connect to electrode **430** and one or more of optional electrodes **449**. A voltage bias can be applied across the capacitive structure **440**, which generates charge trap regions and a conductive path that remain after the voltage bias is removed, thereby providing non-volatile retention, for example as explained in more detail above in connection with FIG. **2**B.

[0101] The MOSFET of optical device **400** can be operated to limit the current flowing through the capacitive structure **440** by applying a gate voltage (V.sub.GS) to gate **404***a* to transition the MOSFET into saturation mode, as described above in connection with FIG. **2**A. Accordingly, in various examples, the MOSFET of optical device **400** can be operated to limit an amount of current flowing through the memristor, so it can control the resistance of the memristor.

[0102] In one example, the memristive behavior may provide for non-volatile phase tuning, which can provide for passband tuning for optical filtering by the optical device **200**. In this case, the optical device may function as an optical filter. For example, waveguide **402** may receive an input optical signal, which can be coupled into the resonator structure **403** based on the coupling coefficient and a resonance frequency of the resonator structure **403**. The resonator structure **440** may exhibit non-volatile retention in which the refractive index of the resonator structure **440** is tuned based on a voltage bias applied across optional electrode **249** and electrode **230**. The change in refractive index causes a phase shift that tunes the resonance frequency of the resonator structure **403**. The tunning can be used to non-volatilely set the resonance frequency, which tunes the coupling of an optical signal in bus waveguide **402** into the resonator structure **402**. This control can be used, for example, to filter undesired wavelengths of light of the optical signal, as well as for encoding/decoding information through frequency modulation (e.g., tuning wavelengths on the optical signal output from optical device **400**).

[0103] According to some implementations, one or more of the capacitive structures of optical device **400** can be utilized for detecting optical signals without extracting light out of a coupled waveguide. For example, capacitive structure **410***a* can be implemented to detect optical power of the optical signal input into the bus waveguide **402**, without extracting the optical signal. In this example, as described above in connection with FIG. **2**B, the capacitive structure **410** can function as a photodetector based on mid-bandgap defects that convert absorbed photons (e.g., from a received optical signal) into photo-generated carriers. That is, for example, bus waveguide **402** may contain defect sites (e.g., defect sites **233** of FIG. **2**B) that can cause the generation of free charge carriers relative to the intensity of the optical signal inside the bus waveguide **402**. Changes in the conductance within bus waveguide **402** may cause variations in current which can be utilized to monitor the intensity of an optical signal propagating in bus waveguide **402**, for example, using a feedback or control circuit as described above in connection with FIG. **2**B.

[0104] In an example, optical device **400** can include an additional optional capacitive structure **410***b*. The capacitive structure **410***b* may be substantially similar to capacitive structure **410***a*, thereby comprising a source **416***b*, gate **404***b* formed on bus waveguide **402**, and dielectric **420***b*, which together with drain **424** constitute a second MOSFET. This MOSFET can be driven, for example by applying a voltage bias across electrode **426***b* (via terminal **482***f*) and electrode **438***b* (via terminal **482***e*) similar to the MOSFET formed using the capacitive structure **410***a*, which may induce an optical phase shift on the optical signal output from the optical device **400**. The second MOSFET may also be used to form a second conductive channel (not shown) similar to conductive channel **432**.

[0105] In various example, capacitive structure **410***b* can be implemented to detect optical power of the optical signal output at an output of bus waveguide **402**, without extracting the optical signal. As described above in connection with FIG. **2**B, the bus waveguide **402** of the capacitive structure **410***b* may contain defect sites (e.g., defect sites **233** of FIG. **2**B) that can cause the generation of free charge carriers relative to the intensity of the optical signal inside the bus waveguide **402**, which can be utilized to monitor the intensity of an optical signal propagating at the output of bus waveguide **402**, for example, using a feedback or control circuit as described above in connection with FIG. **2**B.

[0106] FIG. **5** illustrates an example computing component that may be used to operate optical devices in accordance with various embodiments. Referring now to FIG. **5**, computing component **500** may be, for example, a server computer, a controller, or any other similar computing component capable of processing data. In the example implementation of FIG. **5**, the computing component **500** includes a hardware processor **502**, and machine-readable storage medium for **504**.

[0107] Hardware processor **502** may be one or more central processing units (CPUs), semiconductor-based microprocessors, and/or other hardware devices suitable for retrieval and execution of instructions stored in machine-readable storage medium **504**. Hardware processor **502** may fetch, decode, and execute instructions, such as instructions **506**-**510**, to control processes or operations for controlling optical devices and PICs disclosed herein. As an alternative or in addition to retrieving and executing instructions, hardware processor **502** may include one or more electronic circuits that include electronic components for performing the functionality of one or more instructions, such as a field programmable gate array (FPGA), application specific integrated circuit (ASIC), or other electronic circuits.

[0108] A machine-readable storage medium, such as machine-readable storage medium **504**, may be any electronic, magnetic, optical, or other physical storage device that contains or stores executable instructions. Thus, machine-readable storage medium **504** may be, for example, Random Access Memory (RAM), non-volatile RAM (NVRAM), an Electrically Erasable Programmable Read-Only Memory (EEPROM), a storage device, an optical disc, and the like. In some embodiments, machine-readable storage medium **504** may be a non-transitory storage medium, where the term "non-transitory" does not encompass transitory propagating signals. As described in detail below, machine-readable storage medium **504** may be encoded with executable instructions, for example, instructions **506**-**510**.

[0109] Hardware processor **502** may execute instruction **506** to apply a first voltage bias between a drain and a source of a MOSFET. The MOSFET may be formed, for example, in the optical devices described in connection with FIGS. **1**A-**2**B. In these examples, the MOSFET comprises the drain formed in a device layer comprising a first doped region in a first material (e.g., silicon or other Group IV material), the source formed in the device layer comprising a second doped region in the first material, and the gate comprising a second material (e.g., a Group III-V material) formed on the device layer.

[0110] Hardware processor **502** may execute instruction **508** to form a conductive channel between the drain and the source by applying a second voltage bias between the source and a gate of the MOSFET. In this case, for example, the first waveguide can be provided between the gate and the conductive channel

[0111] Hardware processor **502** may execute instruction **510** to tune the phase of the first waveguide based on adjusting at least one of the first voltage bias and the second voltage bias, wherein modulation depth of the tuning is based on the conductive channel. For example, as described above in connection with FIGS. **1**A-**2**B, applying a voltage across the source and drain (e.g., the first voltage bias) can change carrier concentration in a first waveguide that adjusts a phase of the first waveguide. Additionally, in some examples, the modulation depth of the phase shift in the first waveguide can be adjusted (e.g., increased) based on the conductive channel, as described above in connection with FIGS. **1**A-**2**B. Accordingly, by adjusting the second voltage bias the range of the modulation can be adjusted, which may enable increased granularity in the tuning of the phase due to the first voltage.

[0112] In some examples, as described above in connection with FIGS. **3** and **4**, the MOSFET can be coupled to one of a resonator structure and a MZI. For example, as described in connection with FIG. **3**, the MOSFET may be formed in a first capacitive structure (e.g., capacitive structure **210** as shown in FIGS. **2**A-**3**) coupled to a first arm of the MZI (e.g., arm **313** of FIG. **3**), while a second capacitive structure (e.g., capacitive structure **240** as shown in FIGS. **2**A-**3**) us coupled to a second arm (e.g., arm **311**). While, in the example of FIG. **4**, the MOSFET may be formed in a first capacitive structure (e.g., capacitive structure **410** of FIG. **4**) and coupled to a resonator structure (e.g., resonator structure **403** of FIG. **4**) comprised in a second capacitive structure (e.g., capacitive structure **440**). In either case, a phase shift of a waveguide (e.g., waveguide **203**, **303** and/or the resonator structure **403**) of the second capacitive structure can be tuned to a non-volatile phase shift (e.g., non-volatile retention as described above in connection with FIGS. **2**A-**4**) by applying a third voltage bias across the second capacitive structure. The conductive channel formed by the MOSFET can then function as a current compliance that can limit the current flowing through the capacitive structure to below a current compliance threshold and mitigates thermal breakdown. The current compliance threshold may be dependent on the materials (and doping levels) of the device, as well as the third voltage bias applied across the device.

[0113] In some examples, instruction **510** may be executed by hardware processor **502** to adjust the at least one of the first voltage bias and the second voltage bias based on monitoring optical signals without extracting the optical signal from the device. For example, as described above in connection with FIG. **2**B, the first waveguide may contain defect sites, which generate of free charge carriers relative to the intensity of the optical signal. Monitoring the current passing through the MOSFET can then be used to detect optical power. Thus, for example in the case of a the MZI and/or resonator structure described above, an optical power in one waveguide can be monitored to estimate the optical power in another waveguide and the tuning of the voltage bias can be adjusted based on the detected optical power.

[0114] FIG. **6** depicts a block diagram of an example computer system **600** in which various of the embodiments described herein may be implemented. Computer system **600** may be an example implementation of control circuit **150** of FIG. **1** and/or control circuit **250** of FIGS. **2**A and **2**B. The computer system **600** includes a bus **602** or other communication mechanism for communicating information, one or more hardware processors **604** coupled with bus **602** for processing information. Hardware processor(s) **604** may be, for example, one or more general purpose microprocessors.

[0115] The computer system **600** also includes a main memory **606**, such as a random access memory (RAM), cache and/or other dynamic storage devices, coupled to bus **602** for storing information and instructions to be executed by processor **604**. Main memory **606** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **604**. Such instructions, when stored in storage media accessible to processor **604**, render computer system **600** into a special-purpose machine that is customized to perform the operations specified in the instructions.

[0116] The computer system **600** further includes a read only memory (ROM) **608** or other static storage device coupled to bus **602** for storing static information and instructions for processor **604**. A storage device **610**, such as a magnetic disk, optical disk, or USB thumb drive (Flash drive), etc., is provided and coupled to bus **602** for storing information and instructions.

[0117] The computer system **600** may be coupled via bus **602** to a display **612**, such as a liquid crystal display (LCD) (or touch screen), for displaying information to a computer user. An input device **614**, including alphanumeric and other keys, is coupled to bus **602** for communicating information and command selections to processor **604**. Another type of user input device is cursor control **616**, such as a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **604** and for controlling cursor movement on display **612**. In some embodiments, the same direction information and command selections as cursor control may be implemented via receiving touches on a touch screen without a cursor.

[0118] The computing system **600** may include a user interface module to implement a GUI that may be stored in a mass storage device as executable software codes that are executed by the computing device(s). This and other modules may include, by way of example, components, such as software components, object-oriented software components, class components and task components, processes, functions, attributes, procedures, subroutines, segments of program code, drivers, firmware, microcode, circuitry, data, databases, data structures, tables, arrays, and variables.

[0119] In general, the word “component,” “engine,” “system,” “database,” “data store,” and the like, as used herein, can refer to logic embodied in hardware or firmware, or to a collection of software instructions, possibly having entry and exit points, written in a programming language, such as, for example, Java, C or C++. A software component may be compiled and linked into an executable program, installed in a dynamic link library, or may be written in an interpreted programming language such as, for example, BASIC, Perl, or Python. It will be appreciated that software components may be callable from other components or from themselves, and/or may be invoked in response to detected events or interrupts. Software components configured for execution on computing devices may be provided on a computer readable medium, such as a compact disc, digital video disc, flash drive, magnetic disc, or any other tangible medium, or as a digital download (and may be originally stored in a compressed or installable format that requires installation, decompression or decryption prior to execution). Such software code may be stored, partially or fully, on a memory device of the executing computing device, for execution by the computing device. Software instructions may be embedded in firmware, such as an EPROM. It will be further appreciated that hardware components may be comprised of connected logic units, such as gates and flip-flops, and/or may be comprised of programmable units, such as programmable gate arrays or processors.

[0120] The computer system **600** may implement the techniques described herein using customized hard-wired logic, one or more ASICs or FPGAs, firmware and/or program logic which in combination with the computer system causes or programs computer system **600** to be a special-purpose machine. According to one embodiment, the techniques herein are performed by computer system **600** in response to processor(s) **604** executing one or more sequences of one or more instructions contained in main memory **606**. Such instructions may be read into main memory **606** from another storage medium, such as storage device **610**. Execution of the sequences of instructions contained in main memory **606** causes processor(s) **604** to perform the process steps described herein. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions.

[0121] The term “non-transitory media,” and similar terms, as used herein refers to any media that store data and/or instructions that cause a machine to operate in a specific fashion. Such non-transitory media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, optical or magnetic disks, such as storage device **610**. Volatile media includes dynamic memory, such as main memory **606**. Common forms of non-transitory media include, for example, a floppy disk, a flexible disk, hard disk, solid state drive, magnetic tape, or any other magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, any other memory chip or cartridge, and networked versions of the same.

[0122] Non-transitory media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between non-transitory media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise bus **602**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.
[0123] The computer system **600** also includes a network interface **618** (also referred to as a communication interface) coupled to bus **602**. Network interface **618** provides a two-way data communication coupling to one or more network links that are connected to one or more local networks. For example, network interface **618** may be an integrated services digital network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection to a corresponding type of telephone line. As another example, network interface **618** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN (or WAN component to communicated with a WAN). Wireless links may also be implemented. In any such implementation, network interface **618** sends and receives electrical, electromagnetic or optical signals that carry digital data streams representing various types of information.
[0124] A network link typically provides data communication through one or more networks to other data devices. For example, a network link may provide a connection through local network to a host computer or to data equipment operated by an Internet Service Provider (ISP). The ISP in turn provides data communication services through the world wide packet data communication network now commonly referred to as the "Internet." Local network and Internet both use electrical, electromagnetic or optical signals that carry digital data streams. The signals through the various networks and the signals on network link and through network interface **618**, which carry the digital data to and from computer system **600**, are example forms of transmission media.
[0125] The computer system **600** can send messages and receive data, including program code, through the network(s), network link and network interface **618**. In the Internet example, a server might transmit a requested code for an application program through the Internet, the ISP, the local network and the network interface **618**.
[0126] The received code may be executed by processor **604** as it is received, and/or stored in storage device **610**, or other non-volatile storage for later execution.
[0127] Each of the processes, methods, and algorithms described in the preceding sections may be embodied in, and fully or partially automated by, code components executed by one or more computer systems or computer processors comprising computer hardware. The one or more computer systems or computer processors may also operate to support performance of the relevant operations in a "cloud computing" environment or as a "software as a service" (SaaS). The processes and algorithms may be implemented partially or wholly in application-specific circuitry. The various features and processes described above may be used independently of one another, or may be combined in various ways. Different combinations and sub-combinations are intended to fall within the scope of this disclosure, and certain method or process blocks may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks or states relating thereto can be performed in other sequences that are appropriate, or may be performed in parallel, or in some other manner. Blocks or states may be added to or removed from the disclosed example embodiments. The performance of certain of the operations or processes may be distributed among computer systems or computers processors, not only residing within a single machine, but deployed across a number of machines.
[0128] As used herein, a circuit might be implemented utilizing any form of hardware, software, or a combination thereof. For example, one or more processors, controllers, ASICs, PLAS, PALs, CPLDs, FPGAs, logical components, software routines or other mechanisms might be implemented to make up a circuit. In implementation, the various circuits described herein might be implemented as discrete circuits or the functions and features described can be shared in part or

in total among one or more circuits. Even though various features or elements of functionality may be individually described or claimed as separate circuits, these features and functionality can be shared among one or more common circuits, and such description shall not require or imply that separate circuits are required to implement such features or functionality. Where a circuit is implemented in whole or in part using software, such software can be implemented to operate with a computing or processing system capable of carrying out the functionality described with respect thereto, such as computer system **600**.

[0129] As used herein, the term “or” may be construed in either an inclusive or exclusive sense. Moreover, the description of resources, operations, or structures in the singular shall not be read to exclude the plural. Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps.

[0130] Terms and phrases used in this document, and variations thereof, unless otherwise expressly stated, should be construed as open ended as opposed to limiting. Adjectives such as “conventional,” “traditional,” “normal,” “standard,” “known,” and terms of similar meaning should not be construed as limiting the item described to a given time period or to an item available as of a given time, but instead should be read to encompass conventional, traditional, normal, or standard technologies that may be available or known now or at any time in the future. The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to” or other like phrases in some instances shall not be read to mean that the narrower case is intended or required in instances where such broadening phrases may be absent.

Claims

1. An optical device comprising: a substrate; a device layer formed on the substrate and comprising a first material; a metal-oxide-semiconductor field-effect transistor (MOSFET) formed on the substrate, the MOSFET comprising: a drain formed in the device layer comprising a first doped region; a source formed in the device layer comprising a second doped region; a gate comprising a second material formed on the device layer; and a conductive channel formed in the device layer based on a voltage bias applied to the gate; and a first optical waveguide formed between the gate and the conductive channel.

2. The optical device of claim 1, wherein the first and second doped regions comprise a first doping polarity.

3. The optical device of claim 2, wherein the first material is doped with a first doping concentration and the first doping polarity, wherein the first and second doped regions comprise a second doping concentration that is higher than the first doping concentration.

4. The optical device of claim 2, wherein the second material comprises a second doping polarity that is the opposite of the first doping polarity.

5. The optical device of claim 4, wherein the first doping polarity is p-type doping and the second doping polarity is n-type doping.

6. The optical device of claim 1, further comprising: a first trench formed between the first doped region and the first optical waveguide; and a second trench formed between the second doped region and the first optical waveguide.

7. The optical device of claim 1, further comprising: a memristor connected in series to the drain via the conductive channel.

8. The optical device of claim 7, wherein the memristor is formed in the device layer and comprises a third doped region above the conductive channel, the optical device further comprising a third trench between the second doped region and the third doped region.

9. The optical device of claim 1, wherein the second material comprises a Group III-V

semiconductor material, and wherein the optical device further comprises a metal oxide semiconductor (MOS) capacitor comprising the gate and the first optical waveguide.
10. The optical device of claim 1, further comprising: a Mach-Zehnder Interferometer including a first arm and a second arm, wherein the first arm comprises the first optical waveguide and the second arm comprises a second optical waveguide, wherein the first doped region is between the first and second optical waveguides.
11. The optical device of claim 1, further comprising: a resonator structure formed in the device layer and optically coupled to the first optical waveguide, wherein the resonator structure is connected in series to the drain.
12. A method, comprising: applying a first voltage bias between a drain and a source of a metal-oxide-semiconductor field-effect transistor (MOSFET); forming a conductive channel between the drain and the source by applying a second voltage bias between the source and a gate of the MOSFET, wherein a first waveguide is between the gate and the conductive channel; and tuning a phase of the first waveguide based on adjusting at least one of the first voltage bias and the second voltage bias, wherein modulation depth of the tuning is based on the conductive channel.
13. The method of claim 11, wherein the source, the drain, and the first waveguide are formed in a device layer, and wherein the conductive channel is formed in a region of the device layer below the first waveguide.
14. The method of claim 11, further comprising: tuning a phase shift of a second waveguide to a non-volatile phase shift by applying a third voltage bias to a capacitive structure connected in series to the drain of the MOSFET, the capacitive structure comprising the second waveguide; and limiting current through the capacitive structure based on the conductive channel.
15. The method of claim 14, wherein the capacitive structure comprises one of a resonator structure and an arm of a Mach-Zehnder Interferometer.
16. The method of claim 14, wherein tuning the phase shift of the first waveguide further comprises: forming a filamentation layer in the capacitive structure between a second waveguide and semiconductor material of the capacitive structure by applying the third voltage bias; producing charge trap regions in a buried oxide layer based on forming the filamentation layer, the buried oxide layer provided between a substrate and a device layer that comprises the second waveguide; and creating a conductive path in the buried oxide layer by trapping charge within the charge trap regions such that the capacitive structure operates as a resistor, wherein the conductive path causes a change in a phase of the second waveguide.
17. The method of claim 11, further comprising: receiving an input optical signal by the first waveguide, wherein the input optical signal is coupled into a second waveguide; and monitoring an optical power of the optical signal in the first waveguide based on defect sites in the second waveguide; wherein adjusting the at least one of the first voltage bias and the second voltage bias is based on the monitored optical signal.
18. A photonic integrated circuit (PIC) comprising: a device layer comprising a first material; a first capacitive structure comprising a first anode formed in the first material and comprising a first waveguide, a first cathode comprising a second material, and a dielectric layer disposed between the first cathode and the first anode; and a metal-oxide-semiconductor field-effect transistor (MOSFET) comprising: a source and a drain formed in the device layer; and a gate comprising the cathode, wherein a conductive channel is formed in a region of the device layer based a first voltage bias applied to the gate, and wherein the first waveguide is between the gate and conductive channel.
19. The PIC of claim 18, further comprising: a second capacitive structure electrically connected in series to the MOSFET, the second capacitive structure comprising a second anode formed in the first material and comprising a second waveguide, a second cathode comprising the second material, wherein the dielectric layer is between the second cathode and the second anode, wherein the second capacitive structure exhibits a non-volatile change in an index of refraction of the

second waveguide based on a second voltage applied across the second capacitive structure.

20. The PIC of claim 19, wherein the second capacitive structure is formed in one of a resonator structure optically coupled to the first waveguide and an second arm of a Mach-Zehnder Interferometer (MZI) in which the first capacitive structure is formed in a first arm of the MZI.